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Does parking matter? The impact of parking time on last-mile delivery optimization

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ABSTRACT

Parking is a necessary component of traditional last-mile delivery practices, but finding parking can be difficult. Yet, the routing literature largely does not account for the need to find parking. In this paper, we address this challenge of finding parking through the Capacitated Delivery Problem with Parking (CDPP). Unlike other models in the literature, the CDPP accounts for parking time in the objective and minimizes the completion time of the delivery tour. Parking time represents the process of the delivery person searching for a parking spot and then parking the vehicle at the chosen location. When we restrict the customer geography to a complete grid, we identify conditions for when a Traveling Salesman Problem (TSP) solution that parks at each customer is an optimal solution to the CDPP. We then determine when the parking time is large enough for the CDPP optimal solution to differ from this TSP solution. We introduce a heuristic for the CDPP that quickly finds high quality solutions to large instances. Computational experiments show that parking matters in last-mile delivery optimization. The CDPP outperforms industry practice and models in the literature showing the greatest advantage when the parking time is high. This analysis provides immediate ways to improve routing in last-mile delivery.

1. Introduction

A recent empirical study finds that commercial delivery drivers spend on average 5.8 min searching for each parking spot in Seattle. On average, cruising for parking accounts for 28% of the total trip time between parking locations (Dalla Chiara and Goodchild, 2020). Even if delivery drivers practice illegal parking behaviors, such as double parking, it still may take drivers several minutes to park due to concerns of safety, avoiding conflict with traffic, and cooperating with other drivers (Dalla Chiara et al., 2021). A Principal Scientist at one major package express carrier said, "You can't just double park in front of every customer", further supporting that even double parking takes time. Parking is a necessary component of last-mile delivery practices, and finding a parking spot takes time.

Increasing levels of e-commerce further exacerbates challenges from parking as more vehicles will be competing for each parking spot. In New York City, more than 1.5 million packages need to be delivered every day (Haag and Hu, 2019). These volumes are expected to increase 68% by 2045 (Danigelis, 2018). Attempts to improve productivity in last-mile delivery, partially by eliminating the time needed to find parking, often explore the use of new technology, such as drones or autonomous vehicles (Poikonen and Golden, 2020; Simoni et al., 2020a; Reed et al., 2022b). However, it may be many years before the deployment of such technologies. Today, more immediate solutions are needed to address the challenges posed by the need for parking.

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Although there has historically been recognition that parking takes time and adds complexity to routes (Bodin and Levy, 2000), the current last-mile routing literature largely ignores parking time in route planning. This paper demonstrates that solutions to this parking problem require models to more explicitly incorporate parking decisions. Choosing when and where to park the vehicle is a choice made by the driver, often without the aid of decision support (Boysen et al., 2021). Where decision support is available, it is often in the form of a solution to the Traveling Salesman Problem (TSP). For example, a routing algorithm used by UPS includes a TSP algorithm, but gives drivers autonomy in making final routing decisions about where to drive, where to park, and where to walk (Rosenbush and Stevens, 2015).

To capture the trade-offs between walking to serve more customers from the current parking location and driving to find a new parking location, we model last-mile delivery practices using the Capacitated Delivery Problem with Parking (CDPP). The CDPP is the problem of serving a set of customers with a vehicle and delivery person. The delivery person must park the vehicle to service customers on foot. As the first paper in the literature to explore the impact of parking time, we model this problem deterministically by using a fixed parking time to represent the process of the delivery person searching for a parking spot and then parking the vehicle at the chosen location. The carrying capacity of the delivery person may be restricted based on the number, weight, or volume of packages. The delivery person must return to the parked vehicle after making deliveries on foot. Once returned to the vehicle, the delivery person can load more packages or drive to search for a different parking spot. Under the assumption that driving is faster than walking, we show driving a TSP solution and serving each customer individually is optimal if there is no time associated with parking. However, we prove that parking time only needs to be 1.6 min in urban environments to achieve immediate productivity gains by modeling last-mile delivery as a two-echelon routing problem. Where to park and how often to park become important operational decisions that may improve the productivity of the delivery person, opening a rich area of research in last-mile delivery.

The CDPP generalizes models in the current literature in last-mile delivery, and these generalizations make the mixed integer programming (MIP) formulation for the CDPP difficult to solve. To analyze the impact of parking time, we develop new technology to solve the CDPP to optimality. For instances that face computational limitations, we provide a heuristic for the CDPP that finds high quality solutions quickly. This heuristic captures changes in the solution structure from including parking time.

To understand the impact of considering parking time in routing decisions, we benchmark the CDPP with multiple other vehicle routing problems. The simplest comparison is to use the solution to the TSP with respect to driving times and assume the delivery person parks at every customer. When we restrict the customer geography to a complete grid, we identify conditions for when this TSP solution is an optimal solution to the CDPP. Then, we determine when parking time is large enough for the CDPP optimal solution to differ from this TSP solution. For experimental comparisons, we benchmark the CDPP with models that represent current industry practice as well as recent models in the literature.

While the CDPP was introduced as a benchmark in Reed et al. (2022b), this paper is the first to fully analyze and explore the problem. We bring insight as to why parking matters in last-mile delivery in the following ways:

- We analyze the first model that includes parking time in the objective function. We utilize optimal solutions to the CDPP to evaluate the impact of parking time on last-mile delivery optimization.
- To capture the impact of parking time on optimal routing decisions for larger instances, we provide a heuristic to the CDPP that finds high quality solutions quickly.
- By restricting the customer geography to a complete grid, we identify conditions under which following a TSP tour of the customers and parking at each one is an optimal solution to the CDPP as well as the value of parking time that defines change in the structure of an optimal solution to the CDPP.
- We provide valuable insights from computational experiments showing when the consideration of parking time in the model makes the greatest impact across urban to rural customer geographies.

Section 2 reviews the literature specifically addressing the limited work in vehicle routing that considers the need to park the vehicle. Section 3 discusses the service times, assumptions, and the MIP formulation for the CDPP. To solve larger instances, Section 3.3 provides a heuristic for the CDPP that finds high quality solutions quickly. In Section 4, we directly compare the structure of the CDPP solution with a TSP solution that parks at every customer on a complete grid of customers, providing conditions for when the solutions are equivalent and the value of parking time that defines change in the structure of an optimal solution to the CDPP. Then, Section 5 presents experimental results on real-world customer geographies and discusses the impact of considering parking time on the structure of the solution and the completion time of the delivery tour. Conclusions and future work are discussed in Section 6.

2. Literature review

In this section, we focus on the vehicle routing literature that consider the need to park the vehicle and serve customers on foot. We should note that most literature on last-mile delivery focuses on minimizing driving distance to visit all customers and neglects the need to park the vehicle and determine walking tours (Boysen et al., 2021). We begin by summarizing work that introduced the CDPP to the literature and then discuss studies in routing of last-mile delivery that do not explicitly consider parking time in the objective function.

Reed et al. (2022b) introduce the CDPP as a benchmark to the Capacitated Autonomous Vehicle Assisted Delivery Problem (CAVADP). The CAVADP is the problem of serving a set of customers using an autonomous vehicle assisted by a delivery person. Reed et al. (2022b) provide a MIP formulation for the CDPP but offer no additional study of this problem. The CDPP simply plays the role of being a benchmark model to evaluate the impact of autonomous-assisted delivery on a complete grid of customers, a simplified

representation of urban environments. Similarly, Reed et al. (2022a) use the CDPP as a benchmark when exploring the CAVADP on a general graph to represent urban-to-rural settings. In this paper, we focus on the impact of including parking time in the objective function. To do so, we analytically study optimal solutions to the CDPP on a complete grid and identify conditions under which a TSP solution parking at every customer is optimal. Doing so allows us to understand when parking time impacts optimal routing decisions. We also introduce a high-quality heuristic that can quickly solve large instances of the CDPP providing immediate ways to improve routing in last-mile delivery.

Most studies in routing of last-mile delivery focus on the trade-offs of walking and driving for the delivery person and ignore trade-offs between the difficulty to find parking and other routing decisions. As noted in the literature review of Le Colleter et al. (2023), an earlier version of our paper motivated Le Colleter et al. to include parking time in the Park-and-Loop Routing Problem with Parking Selection that optimizes a fleet of vehicles in last-mile delivery. Unlike the CDPP, Le Colleter et al. (2023) provide a MIP formulation that uses a polynomial set of variables and constraints. The MIP is only tested with 10 and 15 customers. Otherwise, Le Colleter et al. (2023) develop a heuristic. The impact of parking extends to applications outside of last-mile delivery. For example, Cabrera et al. (2022) model parking time in the Doubly Open Park-and-Loop Routing Problem that routes technicians to perform on-site utility services. The complexity of including parking in routing decisions also leads Cabrera et al. (2022) to develop a heuristic. In addition to computational experiments that show the impact of parking time on optimal routing decisions, we also analytically identify when optimal solutions differ from the TSP across urban-to-rural customer geographies to further support the use of parking time when making routing decisions.

The closest problem to the CDPP is the two-echelon last-mile delivery system introduced by Martinez-Sykora et al. (2020). In this system, the decisions are the locations where the vehicle will park, the locations visited by the delivery person on foot, and the order of delivery locations in both the driving and walking routes. The carrying capacity of the delivery person restricts the volume and weight of packages in a customer set. Martinez-Sykora et al. (2020) provide a MIP formulation that uses a polynomial set of variables and constraints but require that one node within each customer set be designated as the parking location. A generalization for the CDPP relative to Martinez-Sykora et al. (2020) is the ability to serve multiple customer service sets from the same parking spot, i.e., the delivery person may service multiple walking loops that start and end at the same parking spot to serve different sets of customers. In addition, this paper provides a model formulation for the CDPP that allows the parking locations to differ from customer locations. Martinez-Sykora et al. (2020) include the clustering of customers as a decision in the optimization problem, but do not include parking time in the objective function. Instead, the objective is a weighted sum of the driving time and walking time for the delivery person. This paper uses the objective function of Martinez-Sykora et al. (2020) to benchmark the objective function in the CDPP. Comparing the solutions with the objective value in Martinez-Sykora et al. (2020) to the solutions using an objective value that includes parking time highlights the impact of parking time on optimal routing decisions. Our results show that the weighting of the driving and walking time in the objective is not sufficient to account for parking and in fact performs quite poorly as a proxy. Martinez-Sykora et al. (2020) use a branch-and-cut algorithm to solve instances up to 30 customers. In this paper, we solve larger instances for the CDPP than the instances in Martinez-Sykora et al. (2020).

Nguyên et al. (2019) define a two-level clustered routing problem to distinguish between the driving route and walking routes of the delivery person. Service time to customers is restricted by time windows. The grouping of customers is an input to the model. Two different partitions of customers are considered: one based on observations of drivers from a case study in London and one based on geographical proximity. Each cluster is required to have a parking location within the cluster. The CDPP generalizes this approach by making the partition of customers and the parking locations optimization decisions. In addition, the CDPP allows multiple customer service sets to be visited from the same parking spot. The optimization decisions in Nguyên et al. (2019) are to select the parking location in each cluster, route the vehicle between the parking locations and depot, and route the walking of the delivery person in each cluster. These decisions capture the delivery person's walk back to the parking location but fail to capture the advantages of serving more than one customer set per parking location if the parking time is high. Nguyên et al. (2019) solve the mixed integer programming formulation for the case study in London and observe that optimizing with respect to time windows can reduce total operation time. However, the parking time is not considered in the operation time. We leave the consideration of time windows in the CDPP for future work and focus this paper on the impact that parking time has on the total time of the delivery tour and the structure of the solution.

More generally, the CDPP can be related to the two-echelon routing problem. In the application of last-mile delivery, one echelon refers to the driving route and the second echelon refers to the walking routes of the delivery person. Sluijk et al. (2023) provide a survey on two-echelon routing problems. The Single Truck-and-Trailer Routing Problem with Satellite Depots (STTRPSD) is similar to the CDPP with the trailer representing the vehicle and the truck representing the delivery person. The truck and trailer are routed on a subset of the satellites (i.e., parking locations for the CDPP) and then the customers are visited from the truck (i.e., delivery person) with routes at each satellite. Belenguer et al. (2016) present a branch-and-cut algorithm to solve the STTRPSD. Belenguer et al. (2016) solve all instances up to 50 customers to optimality and solve 100 to 200 customers to an average optimality gap of 3.02%. The test instances consider at most 10 satellites with 25 to 50 customers and at most 20 satellites with 100 to 200 customers. In our case, we consider all customer locations to be available parking spots (i.e., satellite locations) significantly increasing the size of the model. For instances that face computational limitations, we provide a heuristic for the CDPP.

Unlike the STTRPSD and other two-echelon models for last-mile delivery, the CDPP considers a time associated with searching for a parking spot and then parking the vehicle at the chosen location. This feature aligns the CDPP with the Two-Echelon Capacitated Location-Routing Problem (2E-CLRP) where an opening cost is associated with the satellites (i.e., parking locations). Boccia et al. (2011) present three mixed integer programming formulations for the 2E-CLRP. The 2E-CLRP considers two different fleets of vehicles for first-level and second-level trips, connected by the satellites for transshipment operations. Therefore, the 2E-CLRP can

Table 1
Set of parameters in CDPP.

Notation	Description
n	Number of customers
C	Set of customer locations
S	Set of customer service sets
0	Depot
Π	Set of parking locations and depot
p_i	Expected time to search for parking and then park at parking location $i \in \Pi \setminus \{0\}$ (minutes)
q	Capacity of delivery person (number of packages)
J_i	Set of customer sets that include customer i for $i \in C$
I_{ij}	Indicator variable that customer i is in service set σ_j for $i \in C, \sigma_j \in S$

decompose into two different capacitated location routing problems (Contardo et al., 2012). However, in the CDPP, there exists a dependence between the two levels. The vehicle must remain at the parking spot on the first-level route while the delivery person serves potentially multiple second-level routes on foot. In addition, the travel times in the two levels differ and we capture these differences by using real-world data for the driving times and walking times between customers. For the application of last-mile delivery, the CDPP balances the trade-offs of the delivery person walking and driving to find a new parking location, so differing travel times influence the solution structure. Nguyen et al. (2012b) and Nguyen et al. (2012a) propose metaheuristic approaches to solve the 2E-CLRP. The heuristic proposed in this paper may be applicable to other problems that align with the 2E-CLRP where there is a dependence between the two echelons.

Another line of research uses simulation to model parking availability and the impacts on commercial vehicle parking behavior (Lopez et al., 2019; Nourinejad et al., 2014). Figliozzi and Tipagornwong (2017) combine queuing and logistical models to model parking availability but use continuous approximation models to estimate routing constraints. These papers show that parking has an impact on operations, but still do not consider parking in the routing optimization. In this paper, we argue parking time makes a significant impact on the routing optimization.

3. CDPP

In this section, we provide details on the CDPP and the related MIP formulation. Section 3.1 defines the problem and provides notation for the parameters and service times. Then, Section 3.2 provides the MIP formulation. The CDPP generalizes models in the current literature in last-mile delivery, and these generalizations make the MIP formulation for the CDPP difficult to solve. To analyze solutions to the CDPP for larger instances, Section 3.3 provides a heuristic.

3.1. Problem description and notation

The CDPP serves a set of n customers in a set C by a delivery person with a vehicle. The delivery person and vehicle start and end the tour at the depot, denoted as 0. The delivery person must park the vehicle to service customers on foot. Let Π be the set of parking locations and depot. The time to search for parking and then park at parking location $i \in \Pi \setminus \{0\}$ is p_i minutes. Once parked, the delivery person services a set of customers on foot. After servicing a customer set on foot, the delivery person returns to the parked vehicle. The delivery person can serve another set of customers from this parking spot or move to a different parking spot (e.g. $k \in \Pi$) incurring a parking time of p_k minutes.

Each customer requires a single package delivery. Multiple customer nodes at the same customer location may represent a single customer ordering multiple packages. Let S be the set of potential customer service sets. A customer service set $\sigma_j \in S$ is a subset of the customers that will be served in a walking route, i.e., $\sigma_j \subset C$. We also refer to customer service sets as service sets. The capacity of the delivery person q restricts the number of packages in each customer service set, i.e., $|\sigma_j| \leq q$ for all sets $\sigma_j \in S$. If needed, similar capacity limits relative to package weight or volume may be imposed when generating potential customer service sets. For each $i \in C$, let $J_i = \{\sigma_j \in S | i \in \sigma_j\}$ be the set of customer service sets that includes customer i. Define $I_{ij} = 1$ if $\sigma_j \in J_i$ for all $i \in C$ and $\sigma_j \in S$, and 0 otherwise. Table 1 summarizes the parameters for the CDPP.

The service times require more detail in their definition. Table 2 summarizes the service times for the CDPP. Let $D(i,k)$ be the time to drive between locations i and k for $i,k \in \Pi$. We assume driving times satisfy the triangle inequality. Let d_{ik} be the time to drive from location i to location k and park at k. Then, $d_{ik} = D(i,k) + p_k$ for $i,k \in \Pi$ such that $i \neq k$. In the case where $k = 0$ (i.e., the return to the depot), the vehicle does not need to search for parking and $d_{i0} = D(i,0)$ for all $i \in \Pi \setminus \{0\}$.

Let $W(i,k)$ be the time to walk between locations i and k for $i,k \in C \cup \Pi \setminus \{0\}$. We assume walking times satisfy the triangle inequality. Let w_{ij} be the shortest walking time to service set σ_j when parked at parking location i for $i \in \Pi \setminus \{0\}, \sigma_j \in S$. This walking time is the shortest walk from parking location i to the first customer to be served in set σ_j, the walk between customers in σ_j, and the walk back to parking location i where the vehicle is parked. For the pair $(i, \sigma_j) \in \Pi \setminus \{0\} \times S$, let $(c_1, c_2, \ldots, c_{|\sigma_j|})$ be an optimal order to serve σ_j when parked at i. Then, $w_{ij} = W(i,c_1) + W(c_1,c_2) + \cdots + W(c_{|\sigma_j|-1}, c_{|\sigma_j|}) + W(c_{|\sigma_j|}, i)$.

Let f_j be the time to load package(s) to service set σ_j. We consider a loading time linearly dependent on the number of packages in the service set, i.e., $f_j = f \cdot |\sigma_j|$ for some $f \geq 0$. Therefore, the total loading time in the delivery tour is a constant nf, and the solution to the CDPP is equivalent to the solution when $f_j \equiv 0$ or $f = 0$.

Table 2
Set of service times in CDPP.

Notation	Description
$D(i,k)$	Time to drive from i to k for $i,k \in \Pi$ (min)
d_{ik}	Time to drive from i to k and park at k for $i \in \Pi, k \in \Pi \setminus \{0\}$ such that $i \neq k$ (min)
d_{i0}	Time to drive from parking location i to depot for $i \in \Pi \setminus \{0\}$ (min)
$W(i,k)$	Time to walk from i to k for $i,k \in C \cup \Pi \setminus \{0\}$ (min)
w_{ij}	Time to walk and serve set σ_j while parked at location i for $i \in \Pi \setminus \{0\}$, $\sigma_j \in S$ (min)
f	Time to load one package (min)
f_j	Time to load packages for customer set σ_j for $\sigma_j \in S$ (min)

Table 3
Set of decision variables in CDPP.

Notation	Description
x_{ik}	$x_{ik} = 1$ if the vehicle drives from i to k and parks at parking location k (if $k \neq 0$) for $i,k \in \Pi$ such that $i \neq k$
y_{ij}	$y_{ij} = 1$ if the delivery person is parked at parking location i and serves set σ_j for $i \in \Pi \setminus \{0\}$ and $\sigma_j \in S$
v_{ik}	Flow of packages from location i to location k for $i \in \Pi$ and $k \in \Pi \setminus \{0\}$ such that $i \neq k$

3.2. MIP formulation

Reed et al. (2022b) introduce a variant of the CDPP presented here and provide a MIP formulation. Reed et al. (2022b) take $f_j \equiv g$ for some constant $g \geq 0$. For the purposes of clarity and updated notation, we provide the general formulation here. We also adapt single commodity subtour elimination constraints, as opposed to the adapted MTZ subtour elimination constraints presented in Reed et al. (2022b). Table 3 summarizes the decision variables for the model.

$$\min \sum_{i \in \Pi} \sum_{k \in \Pi \setminus \{i\}} x_{ik} d_{ik} + \sum_{i \in \Pi \setminus \{0\}} \sum_{\sigma_j \in S} y_{ij}(w_{ij} + f_j) \tag{1}$$

$$\text{s.t.} \sum_{i \in \Pi \setminus \{0\}} x_{0i} = 1 \tag{2}$$

$$\sum_{i \in \Pi \setminus \{0\}} x_{i0} = 1 \tag{3}$$

$$\sum_{k \in \Pi \setminus \{0\}} \sum_{\sigma_j \in J_i} y_{kj} = 1 \qquad \forall i \in C \tag{4}$$

$$\sum_{k \in \Pi \setminus \{i\}} x_{ki} = \sum_{k \in \Pi \setminus \{i\}} x_{ik} \qquad \forall i \in \Pi \setminus \{0\} \tag{5}$$

$$y_{ij} \leq \sum_{k \in \Pi \setminus \{i\}} x_{ki} \qquad \forall i \in \Pi \setminus \{0\}, \sigma_j \in S \tag{6}$$

$$\sum_{i \in \Pi \setminus \{0\}} v_{0i} = n \tag{7}$$

$$v_{ik} \leq n \cdot x_{ik} \qquad \forall i \in \Pi, k \in \Pi \setminus \{0\} | i \neq k \tag{8}$$

$$\sum_{k \in \Pi \setminus \{i\}} v_{ki} - \sum_{k \in \Pi \setminus \{0,i\}} v_{ik} = \sum_{\sigma_j \in S} |\sigma_j| y_{ij} \qquad \forall i \in \Pi \tag{9}$$

$$v_{ik} \in \mathbb{Z}_+ \qquad \forall i \in \Pi, k \in \Pi \setminus \{0\} | i \neq k \tag{10}$$

$$x_{ik} \in \{0,1\} \qquad \forall i,k \in \Pi | i \neq k \tag{11}$$

$$y_{ij} \in \{0,1\} \qquad \forall i \in \Pi \setminus \{0\}, \sigma_j \in S \tag{12}$$

The objective function in Eq. (1) minimizes the completion time of the delivery tour. The first term includes the driving time and parking time. The second term is the walking and service time for the delivery person. Constraints (2) and (3) require the vehicle to leave from the depot and return to the depot, respectively. Constraints (4) require that each customer is served in a service set. When the delivery person parks at a parking location, Constraints (5) ensure that the vehicle leaves that parking location. Given that the delivery person services set σ_j when parked at parking location i (i.e., $y_{ij} = 1$), Constraints (6) require the vehicle to drive to and park at location i. Constraints (7)–(10) provide the adapted single commodity subtour elimination constraints. Constraints (7) and (8) ensure n packages flow through the network. While at parking location i, the flow should change by the number of customers in all service sets served from parking spot i. Constraints (9) capture the change in flow. The integer constraints on the v_{ik} variables are given in Constraints (10). Finally, the binary constraints on variables x_{ik} and y_{ij} are given in Constraints (11) and (12), respectively.

To solve the CDPP to optimality, we exploit structure in optimal solutions to identify valid inequalities that raise the lower bound of the MIP and variable reduction techniques that reduce the large number of variables present in the MIP formulation. Appendix A

provides these model improvements that allow us to efficiently solve problems with $n = 50$ customers and $q = 3$ packages. Problems in the related literature solve less than $n = 50$ customers exactly (Martinez-Sykora et al., 2020; Le Colleter et al., 2023) and last-mile delivery routing data supports the use of $q = 3$ packages as a base case (Allen et al., 2018; Merchan et al., 2022). If we want to solve instances with more customers or larger capacities for the delivery person, we face computational limitations due to the growth in the size of the model. To allow us to provide insight on larger instances in our experimental results, Section 3.3 provides a heuristic for the CDPP.

3.3. Heuristic method for the CDPP

To understand the impact of parking time on routing decisions for larger instances, we provide a heuristic for the CDPP that finds high quality solutions quickly. The proposed two-echelon location-routing heuristic decomposes the decisions of the CDPP into two echelons. Section 3.3.1 describes the first echelon, defined as the Parking Assignment and Routing Problem (PA-R), where the customers are assigned to parking locations and the route of the vehicle between these parking locations is defined. For each parking spot, Section 3.3.2 describes the second echelon, defined as the Service Set Assignment Problem (SSA), which determines how to optimally partition the customers in service sets given the parking spots, therefore, defining the walking paths of the delivery person.

The two-echelon location-routing heuristic finds high quality solutions quickly allowing us to extend our discussion on the impact of parking time to instances with more customers and a larger capacity for the delivery person. Appendix A provides detailed results on the heuristic solutions and the quality of these solutions for the experimental design in Section 5.1. For $n = 50$ customers, the heuristic solution is on average within 5.5% of the CDPP optimal value for $q = 1$ to 4 packages. With the largest instance being $n = 100$ customers and $q = 6$ packages, the average runtime for the two-echelon location routing is at most 0.7 min across all customer environments (urban, suburban, and rural) and capacities for the delivery person.

3.3.1. Parking assignment and routing problem

The PA-R determines where to park the vehicle, the assignment of customers to parking locations, and the route of the vehicle between these parking locations. We further decompose the PA-R into two MIPs.

First, we determine where to park the vehicle and assign customers to parking locations. Let $\hat{p}_i = 1$ if the delivery person parks at parking spot i for $i \in \Pi \setminus \{0\}$, 0 otherwise. Let $\hat{a}_{ik} = 1$ if customer k is assigned to parking spot i for $i \in \Pi \setminus \{0\}$ and $k \in C$, 0 otherwise. All service times used in the MIPs of this section are defined in Section 3.1. The cost of opening a parking spot is the parking time p. Therefore, the PA-R captures parking at fewer locations when the parking time is high. Solve the following MIP and denote an optimal solution Z.

$$\min \sum_{i \in \Pi \setminus \{0\}} p \cdot \hat{p}_i + \sum_{i \in \Pi \setminus \{0\}} \sum_{k \in C} W(i,k) \cdot \hat{a}_{ik} \tag{13}$$

$$\text{s.t.} \sum_{i \in \Pi \setminus \{0\}} \hat{a}_{ik} = 1 \qquad \forall k \in C \tag{14}$$

$$\hat{a}_{ik} \le \hat{p}_i \qquad \forall i \in \Pi \setminus \{0\}, k \in C \tag{15}$$

$$\hat{p}_i \le \sum_{k \in C} \hat{a}_{ik} \qquad \forall i \in \Pi \setminus \{0\} \tag{16}$$

$$\hat{p}_i \in \{0,1\} \qquad \forall i \in \Pi \setminus \{0\} \tag{17}$$

$$\hat{a}_{ik} \in \{0,1\} \qquad \forall i \in \Pi \setminus \{0\}, k \in C \tag{18}$$

The objective function in Eq. (13) minimizes a linear combination of the parking time and the assignment of walking (without return walks) to the customers from parking locations. Constraints (14) require that each customer is assigned to a parking spot. Given that a customer is assigned to a parking spot, Constraints (15) require the vehicle to be parked at that parking spot. If no customers are assigned to a parking spot, then Constraints (16) ensure that the parking spot is not opened. Finally, Constraints (17) and (18) give the binary constraints on variables $\hat{p}_i$ and $\hat{a}_{ik}$, respectively.

Now, let $P = \{i \in \Pi \setminus \{0\} | \hat{p}_i = 1 \text{ in } Z\}$ be the parking spots in solution Z. We find the TSP solution of $P \cup 0$ with respect to the driving times, $D(i,k)$, using the standard TSP MIP formulation with single commodity subtour elimination constraints (Gavish and Graves, 1978). This solution provides the route of the vehicle between the parking locations.

3.3.2. Service set assignment problem

For each parking spot, the SSA partitions the customers into customer service sets defining the walking paths of the delivery person. For each $i \in P$, define $K_i = \{k \in C | \hat{a}_{ik} = 1 \text{ in } Z\}$ to be the customers assigned to parking spot i in solution Z. For each $i \in P$, define $S_i \subset S$ to be the potential sets to be served at i, i.e., $S_i = \{\sigma_j \in S | \sigma_j \subseteq K_i\}$. For each $k \in K_i$, let $\hat{J}_k = \{\sigma_j \in J_k | \sigma_j \in S_i\}$ be the service sets in S_i that include customer k. Let $\hat{y}_j = 1$ if σ_j is serviced for $\sigma_j \in S_i$. Solve the following MIP for each parking spot $i \in P$ to determine the service sets.

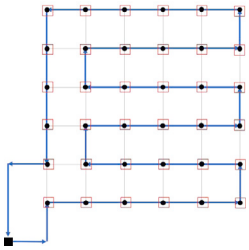

Fig. 1. An example of the TSP solution parking at every customer on a 6 × 6 grid of customers.

$$\min \sum_{\sigma_j \in S_i} w_{ij} \hat{y}_j \tag{19}$$

$$\text{s.t.} \sum_{j \in \hat{J}_k} \hat{y}_j = 1 \qquad \forall k \in K_i \tag{20}$$

$$\hat{y}_j \in \{0, 1\} \qquad \forall j \in S_i \tag{21}$$

The objective function in Eq. (19) minimizes the walking time for the delivery person servicing customers K_i while parked at customer i. Constraints (20) require each customer in K_i to be in a service set. Constraints (21) give the binary constraints on the variables $\hat{y}_j$.

4. When is the TSP solution parking at every customer optimal?

A TSP solution parking at every customer location is a feasible solution to the CDPP when $C \subseteq \Pi$ and often utilized to assist in last-mile delivery routing decisions. Because driving is often faster than walking, this TSP solution may be optimal if the parking time is low, particularly when customers are further apart. However, when customers are close together, accounting for parking time in routing decisions may make it advantageous for the delivery person to consolidate packages into larger customer service sets to reduce the number of times the delivery person searches for parking and parks the vehicle. In this section, we identify conditions under which a TSP solution parking at every customer location is optimal for the CDPP. We also identify when considering parking time in routing decisions is necessary to reduce the completion time of the delivery tour.

We expect the density of customer locations in combination with the parking time to impact the structure of the CDPP solution. To analyze the impact of these features of the problem setting on the CDPP, we consider a complete grid of customers and determine when the parking time is large enough that an optimal solution to the CDPP is no longer this TSP solution. For the purpose of this analysis, we restrict the setting to a $\sqrt{n} \times \sqrt{n}$ complete grid of n customers where $\sqrt{n}$ is even. Let CDPP-grid indicate a problem instance of the CDPP restricted to this grid setting. We assume $\Pi = C \cup 0$ and take $p_k = p$ for all $k \in \Pi \setminus \{0\}$. Let $\hat{d}$ be the time to drive a unit and $\hat{w}$ be the time to walk a unit where $\hat{d} \leq \hat{w}$. The length of a block is $\hat{l}$ units. The capacity of the delivery person is based on the number of packages q assuming that each customer has a single package. Let $(0, 0)$ represent the location of the depot with the bottom left corner of the grid at $(1, 1)$, bottom right corner at $(\sqrt{n}, 1)$, top left corner at $(1, \sqrt{n})$, and top right corner at $(\sqrt{n}, \sqrt{n})$.

We first contemplate the case of a TSP tour through the set of customer locations parking at every customer. Fig. 1 shows an example of this TSP solution on a 6 × 6 grid of customers. The black square indicates the location of the depot and circles represent the customer locations. The solid blue lines represent the path of the vehicle. A red square designates the customer location as a parking spot. Lemma 1 characterizes the completion time of a TSP tour through the grid parking at every customer. The result follows from the analysis in Reed et al. (2022b).

Lemma 1. *Consider a TSP solution where the delivery person parks at every customer location. The objective value for this solution on a $\sqrt{n} \times \sqrt{n}$ complete grid of customers when $\sqrt{n}$ is even is*

$$(2 \cdot MinDistance + n)\hat{d}\hat{l} + nf + np \tag{22}$$

where $MinDistance$ is the minimum distance (in blocks) between the depot and a customer on the grid.

Next, we identify nonzero parking times where an optimal solution to the CDPP-grid is this TSP solution parking at every customer location. We also determine when p becomes large enough that this TSP solution is not optimal. Claim 1 summarizes these results when $q \leq 2$. This and all other proofs can be found in Appendix A.

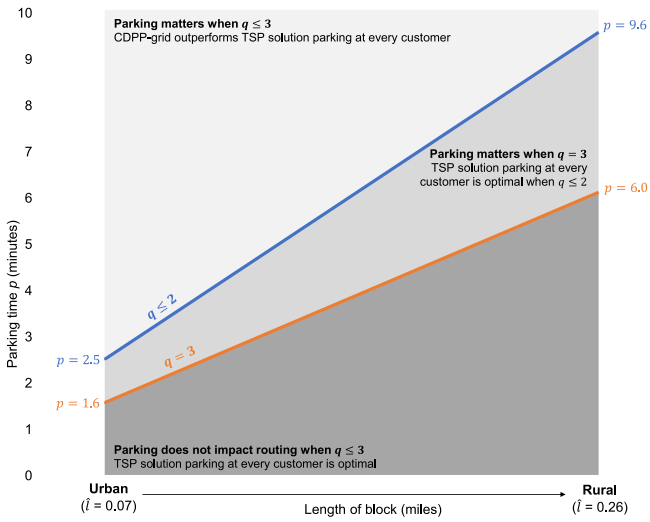

Fig. 2. The value of parking time p (minutes) that determines the optimality of the TSP solution parking at every customer location for the CDPP-grid.

Claim 1. *Assume $q \leq 2$. Then,*

(a) if $p \leq \hat{l}(2\hat{w} - \hat{d})$, then an optimal solution to the CDPP-grid is a TSP solution parking at every customer;
(b) if $p > \hat{l}(2\hat{w} - \hat{d})$, then a TSP solution parking at every customer is not an optimal solution to the CDPP-grid.

Increasing the capacity of the delivery person to $q = 3$ packages allows for gains from consolidating customers into service sets at lower parking times than identified in Claim 1. Claim 1 finds a threshold of $p = \hat{l}(2\hat{w} - \hat{d})$ for the solution structure of the CDPP to change. When $q = 3$, Claim 2 reduces this threshold to $p = \hat{l}(\frac{4}{3}\hat{w} - \hat{d})$.

Claim 2. *Assume $q = 3$. Then,*

(a) if $p \leq \hat{l}(\frac{4}{3}\hat{w} - \hat{d})$, then an optimal solution to the CDPP-grid is a TSP solution parking at every customer;
(b) if $p > \hat{l}(\frac{4}{3}\hat{w} - \hat{d})$, then a TSP solution parking at every customer is not an optimal solution to the CDPP-grid.

Now, we discuss the implications of Claims 1 and 2 on optimal routing decision for last-mile delivery. In both claims, the value of $\hat{l}$, and thus the density of customers, plays a crucial role in determining the structure of the optimal solution to the CDPP-grid. For this analysis, we consider grids representing urban-to-rural settings. We estimate the parameters from the instances presented in Reed et al. (2022a) using real-world data to estimate driving and walking speeds. Section 5.1 further describes the data presented in Reed et al. (2022a). In urban environments (e.g. Cook County in Illinois), we estimate $\hat{l} = 0.07$ miles and $\hat{d} = 4.6$ min/mi. In rural environments (e.g. Cumberland County in Illinois), we estimate $\hat{l} = 0.26$ miles and $\hat{d} = 3.8$ min/mi. We estimate $\hat{w} = 20.3$ min/mi across all urban-to-rural settings.

Fig. 2 shows the value of parking time p that defines change in the structure of an optimal solution to the CDPP-grid when $q \leq 2$ and $q = 3$. For this analysis, we make the simplifying assumption that the values of $\hat{l}$ (the proxy for customer density) and driving speed increase linearly across urban-to-rural settings to analyze the critical value of parking time identified in Claims 1 and 2. In urban environments, when customers are closer together, small values for parking time impact the structure of the CDPP-grid solution. If $p > 1.6$ min, a TSP solution parking at every customer location is not an optimal solution to the CDPP-grid when $q = 3$, the base case of our experimental design. In an empirical study, Dalla Chiara and Goodchild (2020) find that delivery drivers spend on average 5.8 min searching for each parking spot in Seattle. This parking time suggests that productivity gains can be achieved in urban environments by considering parking time in routing decisions. In lower density areas, like rural environments, Fig. 2 shows that the parking time must be higher to change the structure of the optimal solution from a TSP solution parking at every customer. Specifically, when $q \leq 3$, Fig. 2 shows that the TSP parking at every customer is optimal in rural environments if $p < 6$ min. In Section 5, our experimental results indicate that a TSP solution parking at every customer best approximates optimal solutions to the CDPP in rural environments.

5. Experimental results

In this section, we explore the impact of including parking time in optimal routing decisions on the structure of the solution and the completion time of the delivery tour. Section 5.1 discusses the experimental design that enables us to explore the impact of parking time in all customer geographies. Section 5.2 introduces benchmarks to the CDPP that reflect current industry practice as

well as recent models in the literature. Section 5.3 summarizes the differences between the CDPP and benchmark solutions in the base case ($n = 50$ customers, location-dependent parking times, $q = 3$ packages, and $f = 2.1$ min). Then, Sections 5.4 and 5.5 discuss the impact of parking time and capacity of the delivery person, respectively, on the structure of the CDPP solution. We utilize the two-echelon location-routing heuristic to make insights for larger test instances.

5.1. Experimental design

In this section, we introduce the test instances and experimental design. The mixed integer programming model for the CDPP is implemented in Python 3.7.0 using the Gurobi 9.0.0 solver with a 32 thread count on the University of Iowa's Argon high performance computing cluster (Johnson, 2021).

To explore the impact of parking time in all customer geographies, we use the test instances for the case study of Illinois that represent urban to rural settings as described in Reed et al. (2022a). In particular, we use Cook County, Adams County, and Cumberland County to represent urban, suburban, and rural environments, respectively, based on the classification of the United States Department of Agriculture (USDA, 2013). We evaluate ten instances of $n = 50$ customers for each county. The test instances include real-world data for the driving times and walking times between customers (Reed et al., 2021). A study of last-mile delivery practices in London finds that a delivery route visits on average 72 customers (Allen et al., 2018). To evaluate the impact of parking time on instances of size observed in London and larger, we generate five instances in each of Cook, Adams, and Cumberland counties for $n = 100$ following the procedure of Reed et al. (2022a). All test instances are posted at: https://doi.org/10.25820/data.006124.

We restrict the parking locations of the vehicle to all customer locations (i.e., $\Pi = C \cup \{0\}$.) We assume the vehicle finds parking and parks at the chosen location after a fixed parking time. We vary the parking time in each type of customer geography (i.e., urban, suburban, and rural) as parking poses a greater challenge in urban environments than rural environments. Within each geography, we consider a parking time p independent of the parking location (i.e., for all $k \in \Pi \setminus \{0\}$, $p_k = p$ for some $p \geq 0$.) Doing so allows us to isolate the impact of the parking time on optimal routing decisions in different geographies. Like Reed et al. (2022a), we use $p = 9$ min in Cook County, $p = 5$ min in Adams County, and $p = 1$ min in Cumberland County as our base case. These values reflect location-dependent parking times where a higher parking time is realized in urban environments compared to rural environments. For each county, we also experiment with smaller values of p to understand how including the parking time changes the structure of the solution in the CDPP. For Cook County, we experiment with $p = 0, 3,$ and 6 min. For Adams County, we include experiments with $p = 0$ and 3 min. For Cumberland County, we also consider $p = 0$.

We choose the parameter values for the capacity of the delivery person q and the time to load a package f based on observations in the literature. A study of last-mile delivery in London finds that a delivery person delivers 3 packages on average per stop (Allen et al., 2018). The dataset used in the 2021 Amazon Last Mile Routing Research Challenge shows drivers in United States metropolitan areas deliver on average 238.5 packages at 148.0 stops which translates to an average of 1.6 packages per stop (Merchan et al., 2022). Therefore, we consider a carrying capacity based on the number of packages q and use $q = 3$ packages in our base case. We experiment with capacities of 1 to 6 packages. The average delivery time for a package is estimated to be between 2.5–4.1 min (Allen et al., 2018; Simoni et al., 2020b; Zhang et al., 2018). This time often includes multiple delivery activities. Allen et al. (2018) observe 4.1 min of service time per customer where this estimate includes the time spent unloading the package, walking to the customer, and gaining proof-of-delivery. Nguyên et al. (2019) estimate a one minute walking time allowance and one minute consignee service time. Combining the estimates of Allen et al. (2018) and Nguyên et al. (2019), we assume the time to unload a package is 2.1 min in our base case.

5.2. Benchmarks

We introduce two benchmarks to the CDPP and use these benchmarks to highlight why including parking time matters in routing optimization for last-mile delivery.

5.2.1. Modified TSP

The routing algorithm for UPS provides the driver a solution based on a TSP algorithm (Rosenbush and Stevens, 2015). The driver has autonomy to make the final routing decisions including where the delivery person will walk and where the delivery person will drive. To model this real-life practice, we use a TSP solution to fix the order of service. To reflect choices made by the driver and allow service to customers on foot, we transform this TSP solution to take into account the trade-offs between the parking time, driving time, and walking time.

This benchmark, hereafter called the Modified TSP, is a route-first cluster-second method to optimize the trade-offs of walking, driving, and parking given a fixed customer order. A TSP solution with respect to driving times fixes the order of customer service. We generate potential service sets based on this order and restrict the size of the service set based on the capacity of the delivery person q. In addition, we allow the delivery person to serve multiple customer service sets from the same parking spot while maintaining the order of customer service. We implement this benchmark by restricting the sets of driving variables x_{ik} and service variables y_{ij} in the CDPP. We update the walking service time w_{ij} to maintain the order of service. We use the objective function in Eq. (1) to minimize the completion time of the delivery tour and capture the impact of parking time on the driver's decision making.

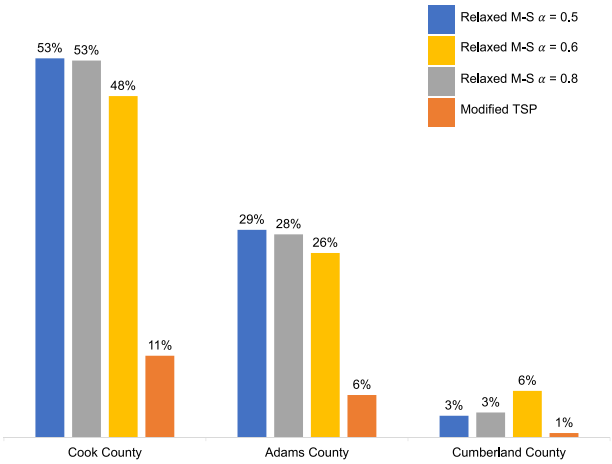

Fig. 3. Average percent reduction in completion time of delivery tours by using CDPP relative to Relaxed M-S with $\alpha = 0.5$, Relaxed M-S with $\alpha = 0.6$, Relaxed M-S with $\alpha = 0.8$, and Modified TSP in the base case.

5.2.2. Relaxed M-S

One of the limitations in the two-echelon models presented by Nguyên et al. (2019) and Martinez-Sykora et al. (2020) is that both require a parking location in every customer set. Unlike Nguyên et al. (2019), Martinez-Sykora et al. (2020) include the clustering of customers as a decision in the optimization problem. To benchmark the CDPP with the literature, we consider a relaxed version of the model by Martinez-Sykora et al. (2020), hereafter called Relaxed M-S, that allows the delivery person to serve multiple customer service sets from the same parking spot. For comparison purposes, we define the carrying capacity of the delivery person in the Relaxed M-S benchmark to be based on the number of packages q.

The key difference between the Relaxed M-S and the CDPP is the objective function. The objective function for the CDPP, in Eq. (1), includes the parking time when evaluating the completion time of the delivery tour. Martinez-Sykora et al. (2020) consider a weighted sum of the driving and walking times for the delivery person. Using the notation of Section 3.1, Eq. (23) presents the objective function in Martinez-Sykora et al. (2020),

$$\alpha \sum_{i \in \Pi} \sum_{k \in \Pi \setminus \{i\}} x_{ik} D(i,k) + (1-\alpha) \sum_{i \in \Pi \setminus \{0\}} \sum_{\sigma_j \in S} y_{ij} w_{ij} \tag{23}$$

where $\alpha \in [0,1]$. We implement the Relaxed M-S benchmark by using Eq. (23) as the objective value in the CDPP. Since we assume a loading time linearly dependent on the number of packages, the total loading time is a constant and added to the objective value in Eq. (23). Similar to Martinez-Sykora et al. (2020), we consider $\alpha \in \{0.6, 0.8\}$ in the Relaxed M-S benchmark. In addition, we consider $\alpha = 0.5$ which gives an equal weighting to driving and walking time. Note that setting $\alpha = 0.5$ provides an equivalent solution to the CDPP with $p = 0$ min. Let v be the optimal value of the Relaxed M-S benchmark (including loading time) and s be the number of times the vehicle parks in the respective solution. Then, the completion time with a solution of this benchmark is $v + sp$ minutes. Comparing the completion times for the CDPP and the Relaxed M-S benchmark highlights the impact of including parking time in the objective function.

5.3. Comparison of CDPP to benchmarks

In this section, we compare the completion time of the delivery tour in the CDPP with the values of the solutions for the benchmarks in the base case ($n = 50$ customers, location-dependent parking times, $q = 3$ packages, and $f = 2.1$ min).

Fig. 3 shows the average percent reduction in the completion time of delivery tours by using the CDPP relative to each benchmark for Cook, Adams, and Cumberland counties. Including parking time reduces the completion time of delivery tours in all counties. The CDPP outperforms models in the literature and industry practice. The Relaxed M-S benchmark generalizes the current models in the literature. Fig. 3 shows the CDPP reduces the completion time up to 53% on average relative to the Relaxed M-S benchmark with $\alpha = 0.5$. We see similar results when $\alpha = 0.6$. Increasing α to 0.8, the CDPP reduces the completion time up to 48% on average. In Cumberland County, the CDPP realizes greater reductions at higher levels of α. Considering the Modified TSP to model real-life practice, the CDPP reduces the completion time up to 11% on average. Insight 1 summarizes this result.

Insight 1. *Parking matters in last-mile delivery optimization. The CDPP outperforms industry practice and models in the literature, highlighting the value of determining the order of service and including parking time in optimal routing decisions.*

The impact of parking differs across customer geographies. The CDPP provides the greatest savings in Cook County, an urban environment with the highest customer density and parking time. Including the parking time in routing decisions via the CDPP

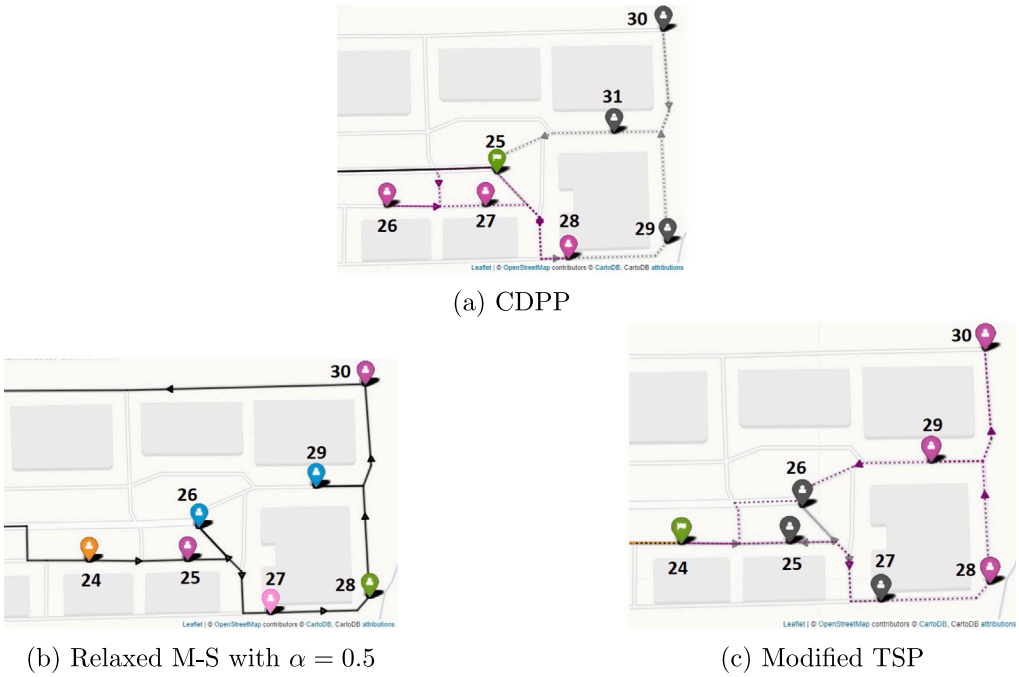

Fig. 4. Solutions to a portion of an instance of Cook County for the (a) CDPP, (b) Relaxed M-S benchmark with $\alpha = 0.5$, and (c) Modified TSP benchmark in the base case.

reduces the completion time of these delivery tours by an average of 53% relative to considering the driving and walking trade-off via the Relaxed M-S benchmark with $\alpha \in \{0.5, 0.6\}$. Further, the delivery person saves an average of 48% and 11% in delivery time relative to the Relaxed M-S with $\alpha = 0.8$ and Modified TSP, respectively. In rural areas where the customer density and parking time is generally low, the CDPP provides a similar completion time in the delivery tour as the Modified TSP. Therefore, using the TSP solution may be sufficient in making routing decisions for rural environments. Increasing the value of α in Eq. (23) makes walking advantageous in the solution to the Relaxed M-S benchmark. However, in rural environments, when the distance between customers is greater, a solution that incentivizes walking adversely affects the realized completion time of the delivery tour. Therefore, we see a higher level of savings for the CDPP when $\alpha = 0.8$ than $\alpha = 0.5$ or 0.6. We further explore this result later in this section. Insight 2 summarizes these results.

Insight 2. *Including parking time in routing optimization for last-mile delivery provides the greatest advantage in urban environments where parking is a challenge. In rural areas, the TSP may be sufficient in making routing decisions.*

Now, we explore how the structure of an optimal solution to the CDPP differs from the solutions of the benchmark problems. Fig. 4 shows the solutions to the (a) CDPP, (b) Relaxed M-S with $\alpha = 0.5$, and (c) Modified TSP for a portion of an instance in Cook County. We focus on a portion of the solution to highlight the local differences in the CDPP and benchmark solutions. The solid black lines indicate the driving path of the vehicle. The dotted lines indicate the walking paths of the delivery person. Each color represents a different service set. The flag icons indicate the parking spots. The numerical label at the customer location indicates the order service within the tour. For example, a numerical label of i indicates that this customer is ith on the tour. Fig. 4a shows an optimal solution for the CDPP in this instance of Cook County. The delivery person serves three customer sets from a single parking spot. Fig. 4b shows the solution for the Relaxed M-S with $\alpha = 0.5$ (i.e., CDPP with $p = 0$). For this instance, the solution to this benchmark is a TSP solution, parking at every customer, and driving between customer locations. However, the parking time in Cook County is 9 min. If the delivery person follows the solution in Fig. 4b, the delivery person spends about an hour looking for parking to serve these eight customers. Fig. 4a shows that even though walking is slower than driving, it is more advantageous to park once and serve all customers on foot. Insight 3 summarizes this observation.

Insight 3. *When parking time is ignored, routing decisions that focus on finding the fastest driving path to service all customers may result in unnecessary time spent searching for parking.*

Recall that the Modified TSP uses a solution to the TSP to fix the order of customer service. Fig. 4c shows the solution of the Modified TSP benchmark. The Modified TSP optimizes the trade-offs between walking, driving, and parking for a fixed service order. Similar to the CDPP solution in Fig. 4a, Fig. 4c shows that the delivery person parks once to avoid high parking times and walks to service customers on foot. However, fixing the order of service to the TSP results in additional walking time for the delivery

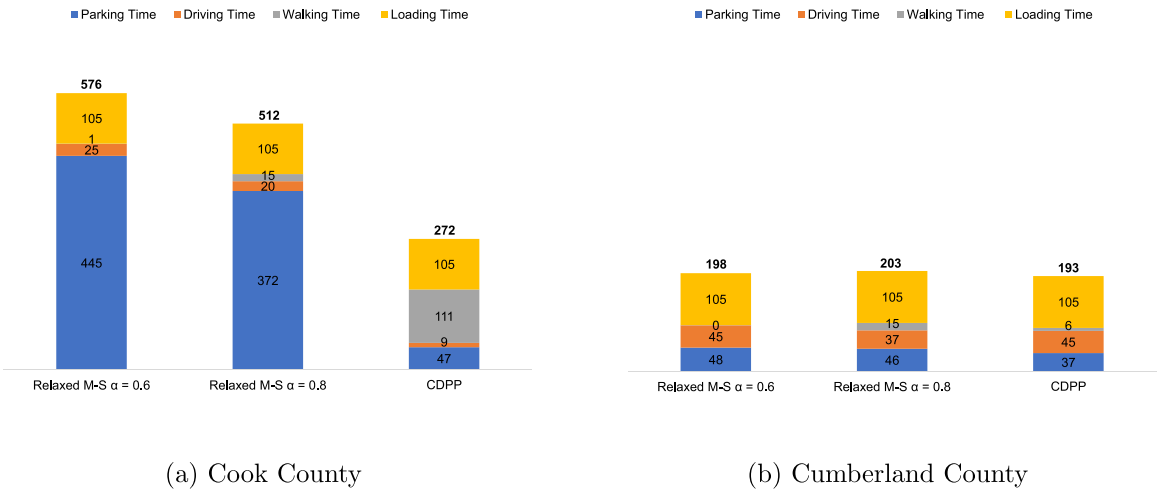

Fig. 5. Average time (minutes) spent in the Relaxed M-S with $\alpha \in \{0.6, 0.8\}$ and CDPP delivery tours parking, driving, walking, and loading packages for (a) Cook County and (b) Cumberland County in the base case.

person relative to the CDPP in Fig. 4a. The CDPP takes into account the parking time when determining where to park as well as the walking path for the delivery person reducing the completion time of the delivery tour relative to the Modified TSP. Insight 4 summarizes this observation.

Insight 4. *Routing in last-mile delivery must include optimizing customer service order to make more efficient choices of parking spot locations and walking paths for the delivery person.*

Finally, we analyze the solution structure of the Relaxed M-S benchmarks to understand the impact of using parking time in the objective function of the CDPP. Fig. 5 shows the average time (in minutes) spent in the Relaxed M-S and CDPP delivery tours parking, driving, walking, and loading packages for Cook and Cumberland counties. Similar to Martinez-Sykora et al. (2020), we consider $\alpha \in \{0.6, 0.8\}$ for this analysis. When $\alpha = 0.6$, the solution structure in both counties relies on the delivery person driving with limited to no walking. A solution that focuses on driving forces the delivery person to search for many parking locations. In Cook County, Fig. 5a shows the delivery person spends on average 77% of the delivery tour in the process of parking. Increasing α to 0.8 makes walking advantageous and reduces the completion time of the delivery tour by 11% on average. However, the process of parking remains a significant portion of the delivery tour. Including the parking time in the objective function further reduces the completion time of the delivery tour by 48% on average. In Cook County, where the customer density is high, the delivery person controls the time spent parking by parking in fewer locations and walking to service customers. When customers are further apart, like Cumberland County, Fig. 5b shows that increasing α to 0.8 increases the completion time of the delivery tour 3% on average. Making walking advantageous in rural environments forces the delivery person to walk between customers that are further apart. The similarity in the solution structures between the Relaxed M-S with $\alpha = 0.6$ and the CDPP support the conclusions of Insight 2 that a solution that relies on driving is sufficient in rural environments. Insight 5 summarizes these results.

Insight 5. *Including parking time in the objective for last-mile delivery routing is critical to achieve optimal trade-offs between driving, walking, and parking.*

5.4. Impact of parking time

In this section, we focus on how the parking time p changes the structure to the CDPP solution for each type of customer geography based on county. Fig. 6 shows the average time (in minutes) spent in optimal CDPP delivery tours parking, driving, walking, and loading packages for Cook County in the base case with various parking times. In urban settings, parking time is expected to be high. We test $p = 0, 3, 6$, and 9 min. Fig. 6 shows that the total time spent in the process of parking remains relatively stable – between 40–47 min on average – when $p > 0$, indicating that the delivery person parks fewer times as the parking time increases. Relatedly, we observe the walking time significantly increases (57 to 111 min on average) as the delivery person must walk further to service more customers. Insight 6 summarizes this observation.

Insight 6. *In urban areas, as parking time increases, the total time spent in the process of parking remains stable. At higher parking times, the delivery person parks fewer times at the expense of significantly increasing walking time.*

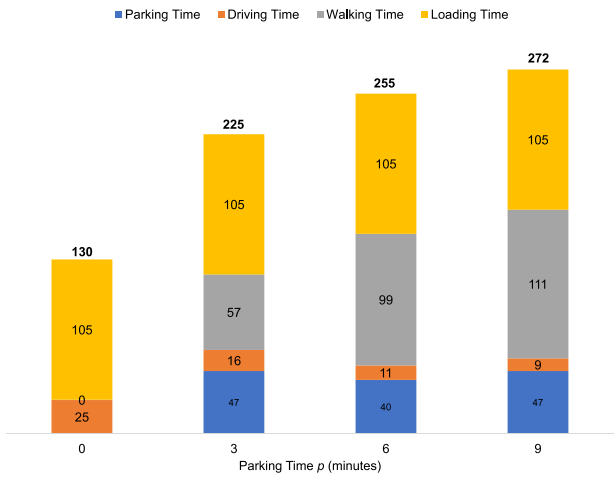

Fig. 6. Average time (minutes) spent in optimal CDPP delivery tours parking, driving, walking, and loading packages for Cook County in the base case with various parking times.

Outside of urban environments, we observe that differences in customer geography reduce the impact of parking time on the structure of the CDPP solution. Fig. 7 shows the average time (in minutes) spent in optimal CDPP delivery tours parking, driving, walking, and loading packages for Adams and Cumberland counties. Recall that Adams and Cumberland counties are classified as suburban and rural areas, respectively. Therefore, we expect customers to be further apart in these counties than in an urban area, like Cook County. When p increases from 0 to 3 min, the completion time of the delivery tour increases by approximately the same amount on average in Cook and Adams counties (95 min in Cook County and 98 min in Adams County). However, the impact on the structure of the solution differs. When p increases from 0 to 3 min in Cook County, Fig. 6 shows the average time spent in the process of parking increases 47 min accounting for 49% of the increase in the completion time of the delivery tour. The remaining 51% of the increase reflects changes in the solution structure, i.e. less driving time and more walking time for the delivery person. In Adams County, when p increases from 0 to 3 min, Fig. 7a shows that the average time spent in the process of parking increases 65 min. In this case, the total parking time accounts for 66% of the increase in the completion time of the delivery tour. Similar to Cook County, when p increases from 0 to 3 min, Fig. 7a shows driving time decreases and walking time increases, but these changes in the solution structure only account for 33% of the increase in the completion time of the delivery tour. Therefore, the increase in the parking time has less impact on the solution structure in Adams County than Cook County. The differences in how the solution changes between Cook and Adams counties reflect that the CDPP solution cannot trade off more walking at a higher parking time in a county, such as Adams, where the customers are relatively further apart. Insight 7 summarizes these results.

Insight 7. *Differences in customer geography influence the significance of including the parking time in routing decisions. Increasing the parking time outside of urban environments has less impact on the solution structure than in urban environments.*

In rural environments, like Cumberland County, we test $p = 0$ and 1 min as we expect parking times to be low. Since customers are likely further apart than urban and suburban environments, we expect the delivery person to spend more time driving between customers than walking. Fig. 7b shows that there exists customer locations close enough together such that it is advantageous to walk to them even when $p = 1$ min. However, the driving time remains the same on average indicating that the increase in the parking time does not have a significant effect on routing decisions. These observations support the conclusion of Insight 2 that the TSP driving to all customers may be sufficient in making routing decisions for rural environments.

Insight 7 concludes that increasing parking time has the greatest impact in urban environments. Now, we show the two-echelon location-routing heuristic captures similar changes to the structure of the solutions as parking time changes for larger n in urban environments. In optimal solutions to the CDPP for $n = 50$ in Cook County, Fig. 6 shows that as parking time increases, the delivery person parks fewer times at the expense of significantly increasing walking time. Fig. 8 shows the results for heuristic solutions on $n = 100$ instances in Cook County using the two-echelon location-routing heuristic. The figure shows the average time of the delivery tour (in minutes) spent in the heuristic solutions with various parking times. Again, we observe greater change in the walking time for the delivery person relative to changes in parking time. Thus, it remains advantageous to park fewer times and walk further to deliver to customers. This analysis supports the use of the two-echelon location-routing heuristic to realize the advantages of including the parking time in routing last-mile delivery for larger instances.

5.5. Impact of the capacity of delivery person

In this section, we discuss the impact of the capacity of the delivery person q on reducing the completion time of the delivery tour in different customer geographies. Fig. 9 shows the average objective value of the CDPP for Cook, Adams, and Cumberland counties

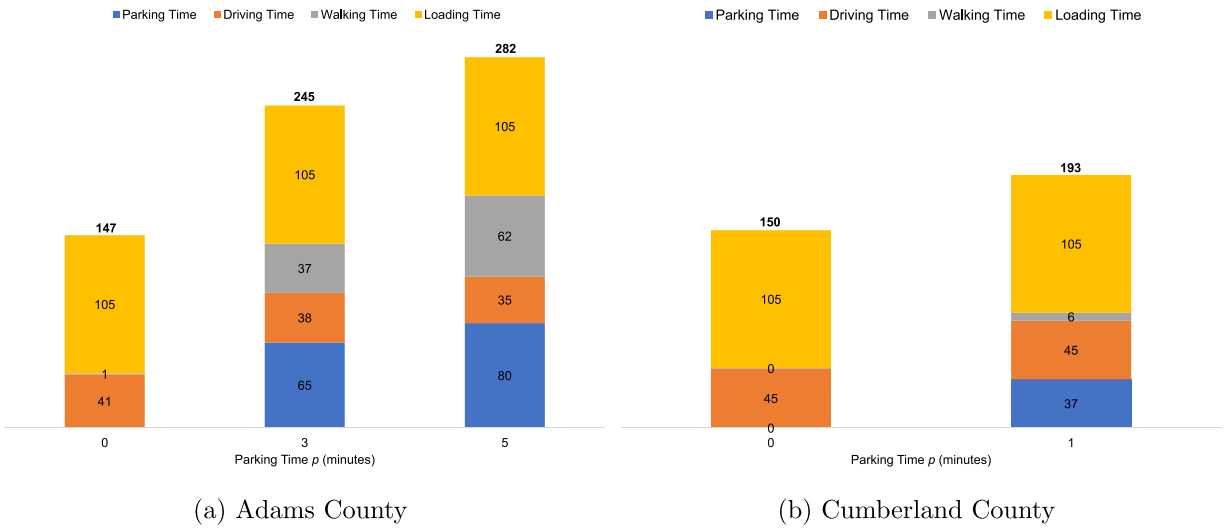

Fig. 7. Average time (minutes) spent in optimal CDPP delivery tours parking, driving, walking, and loading packages for (a) Adams County and (b) Cumberland County in the base case with various parking times.

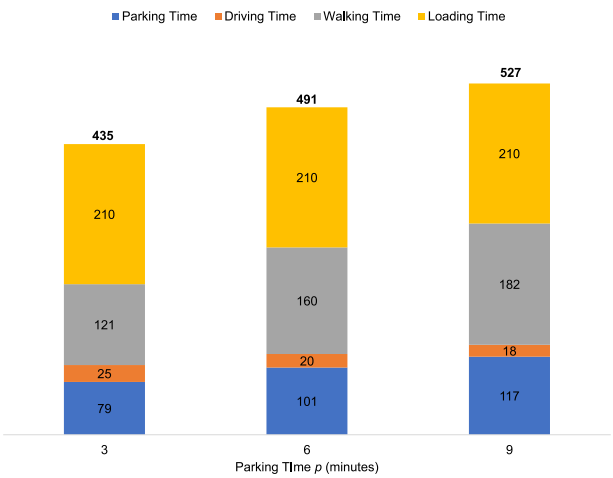

Fig. 8. Average time (minutes) spent parking, driving, walking, and loading packages in solutions to the two-echelon location-routing heuristic for Cook county when $n = 100$ in the base case with various parking times.

in the base case ($n = 50$ customers) with varying capacities. A textured bar indicates the use of the two-echelon location-routing heuristic for the analysis. In urban areas, like Cook County, Fig. 9 shows increasing the capacity from $q = 1$ to 4 packages reduces the completion time of the delivery tour by 20% on average. Increases in the completion time when $q = 5$ and 6 packages are due to the use of the heuristic solution in the analysis. We observe a smaller impact in Adams County with an average reduction of 6% when increasing from $q = 1$ to 4 packages. In rural areas, like Cumberland County, we observe little difference between the solution for $q = 1$ and 4 packages. This observation supports Insight 2 showing that serving customers individually may be sufficient for routing decisions in rural areas. For all counties, we observe marginally decreasing reductions in the delivery tour when increasing capacity. Fig. 10 shows the average objective value of the CDPP for each county when $n = 100$ with varying capacities. Note that the CDPP MIP for 1 out of the 5 instances in Cumberland County for $n = 100$ and $q = 1$ and 2 solved to an optimality gap of 1.2%. Fig. 10 shows similar results with respect to changes in capacity of the delivery person when $n = 100$ customers which again supports the use of the two-echelon location routing heuristic to capture changes in the solution with respect to capacity. Insight 8 summarizes these results.

Insight 8. *Increasing the capacity of the delivery person is more advantageous outside of rural areas. We observe diminishing marginal returns with each addition to driver capacity.*

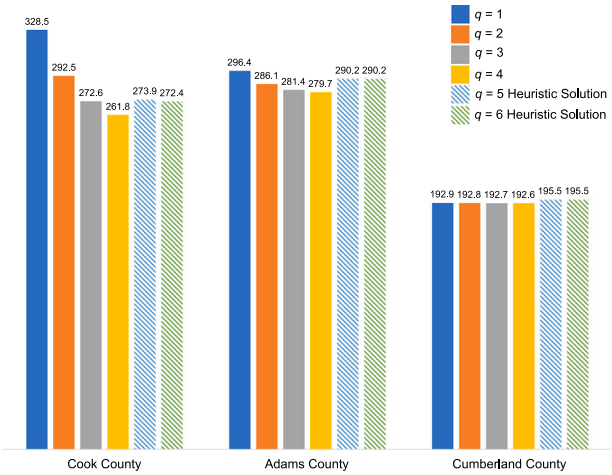

Fig. 9. Average optimal delivery tour completion time (minutes) of the CDPP (or heuristic) for Cook, Adams, and Cumberland Counties in the base case ($n = 50$ customers) with varying capacities.

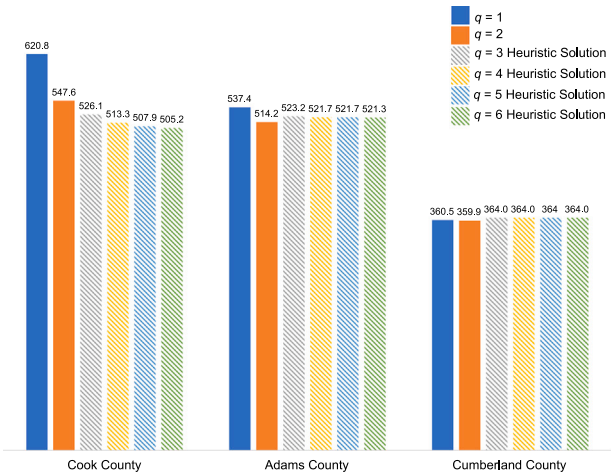

Fig. 10. Average optimal delivery tour completion time (minutes) of the CDPP (or heuristic) for Cook, Adams, and Cumberland Counties when $n = 100$ customers with varying capacities.

6. Conclusions and future work

Yes, parking matters. Including parking time in the objective for routing optimization for last-mile delivery may significantly reduce the completion time of the delivery tour. Routing to minimize distance traveled may not minimize route time as the driver could spend unnecessary time parking the vehicle. Using the CDPP for routing decisions provides the greatest advantage in urban environments where parking is a challenge and often time consuming. When parking is a challenge, the CDPP solution recommends that the delivery person park in fewer locations and serve multiple customer service sets from the same parking spot. This decision balances the trade-offs of walking to service more customers from the current parking location and driving to find a new parking location. However, in rural environments where parking is more readily available, a solution that parks at all customers serving each customer individually may be sufficient.

These insights are reflected in our analytical results that determine when the parking time becomes large enough that the TSP solution parking at every customer location is not an optimal solution to the CDPP-grid. These results show that in urban environments, where customers are close together, small parking times impact the structure of the optimal solution in the CDPP-grid. A TSP solution that parks at every customer best approximates optimal solutions to the CDPP-grid in rural environments where customers are further apart.

This paper introduces technology for solving instances with $n = 50$ customers and $q = 3$ packages for the CDPP efficiently. However, further work needs to be done to control the growth in the number of variables in the model, particularly the service variables y_{ij} which grows both in the number of customers as well as the number of potential service sets. An alternative formulation

of the CDPP using traditional subtour elimination constraints or column generation may improve computational performance. In addition, the model improvements in this paper may inform ways to improve the MIP formulations in Martinez-Sykora et al. (2020) and Le Colleter et al. (2023) that use a polynomial set of variables and constraints. A generalization of the branch-and-cut algorithm from Martinez-Sykora et al. (2020) that allows multiple customer service sets from the same parking spot may also improve computational performance. For instances where the CDPP formulation becomes intractably large, we propose a two-echelon location-routing heuristic and show that this heuristic finds high quality solutions quickly. These heuristic solutions outperform other traditional last-mile delivery models providing an immediate improvement to industry practice and models in the literature.

This work provides immediate ways to improve routing a single vehicle for last-mile delivery. Future work includes considering the impact of parking under additional delivery constraints, such as customer time windows and additional service times, as well as the consideration of a fleet of vehicles. Further, the impact of parking may differ based on the density of customer locations in the same type of customer geography. In the experimental design, we assume that the parking locations are restricted to customer locations. However, it may be the case, as studied in Le Colleter et al. (2023), that the vehicle cannot park at every customer and/or there exists parking locations, such as loading zones, outside of customer locations that should be considered. Future work includes the analysis of further restricting or increasing parking locations in the CDPP.

We model last-mile delivery in a deterministic framework to build insights on the impact that parking time has on routing decisions. However, we know parking is stochastic and can vary by location and time of day. In Seattle, Dalla Chiara and Goodchild (2020) conclude that the cruise time for parking decreases when more curb-space is allocated to commercial loading zones and paid parking and increases when more curb-space is allocated to bus zones. A survey of 16 drivers in New York City finds that the search time for parking ranges from 3 min in Brooklyn to 60 min in Midtown East (Holguín-Veras et al., 2016). Future work includes the analysis of spatial changes in parking time. In addition, future work includes generalizing the CDPP to consider temporal changes in parking time as well as the impact of parking in a stochastic framework. In particular, we may utilize insights from the CDPP to model parking in a dynamic setting where the driver may choose an alternate parking spot based on observed parking availability.

An understanding of the availability of parking locations and its impact on delivery practices also benefits urban planning efforts. Currently, many urban areas are experimenting with loading zone reservations and pricing schemes for delivery vehicles (Shaver, 2019; Balik et al., 2016). For example, a pilot program in Aspen, Colorado, allows commercial drivers to reserve and pay for the use of "Smart Zones" with a mobile app (Sackariason, 2020). Insights into how parking impacts routing in last-mile delivery may allow for better placement of loading zones and better pricing schemes to incentivize drivers and yield better curbside management.

CRediT authorship contribution statement

Sara Reed: Conceptualization, Methodology, Software, Writing – original draft, Visualization. **Ann Melissa Campbell:** Conceptualization, Resources, Writing – review & editing. **Barrett W. Thomas:** Conceptualization, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.tre.2023.103391.

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